UCC501: GEOTECHNICAL ENGINEERING-I

Prerequisite(s): NIL	**L**	**T**	**P**	**Cr**
	3	**1**	**2**	**4.5**

Course Objectives: This subject aims to develop an understanding of soil as civil engineering material and to introduce the students about the basic concepts and principles of soil mechanics. Further they will be introduced to the concepts of compaction, consolidation and determination of shear strength of soil.

Introduction: Soil formation, various soil types.

Phase relationships: Index properties, sieve & hydrometer analysis, Atterberg's limits, sensitivity, thixotropy, and plasticity charts. Determination of engineering properties of soil. Indian standard and Unified classification systems of soils.

Clay Mineralogy: Introduction to Clay minerals their characteristics. Soil structure.

Seepage and Permeability: Darcy's law, validity of Darcy's Law, seepage velocity, factors affecting permeability, Laboratory and field determination of permeability. Flow net and its properties, Laplace equation, methods of drawing flownet, seepage through earth dams, exit gradient and seepage pressures, phenomenon of piping and heaving, filters. Anisotropy, Permeability of layered soils.

Effective Stress Principle Capillarity, types of head, seepage forces, quick sand condition, and critical hydraulic gradient.

Compaction: Compaction tests as per IS code, OMC, factors affecting compaction, control of compaction, field compaction equipment and their suitability.

Stresses in Soils: Stresses beneath various loaded areas, Boussinesq and Westergarrd's formulae, pressure bulbs, Newmark's chart. Approximate methods.

Consolidation: Terzaghi's theory, time rate of consolidation, consolidation test, Compressibility & Coefficient of Consolidation, NC, OC soils, determination of pre-consolidation pressure, settlement analysis, secondary consolidation.

Shear Strength: Definition, Mohr's stress circle, Mohr-Columb strength theory, direct, triaxial, unconfined and vane shear tests. Drainage conditions, Concept of pore pressure coefficients, shear characteristics of normally consolidated, over consolidated clays and dense and loose sands, Dilatancy, residual strength.

Laboratory Work:

The students will be introduced to Index and Engineering properties of soils to complement the theory component of the course by performing experiments. They will perform related experiments as per BIS specifications.

1. Determination of field density by Core cutter & Sand replacement method
2. Grain size Analysis by Mechanical method.
3. Determination of Relative density of coarse grained soils in dry and saturated conditions.
4. Determination of Specific Gravity by Pycnometer.
5. Determination of Atterberg's limits.
6. Determination of Permeability by constant head & variable head permeameter.
7. Determination of Coefficient of Consolidation by Consolidation Test.
8. Determination of OMC and MDD by IS standard Compaction test.
9. Determination of Unconfined compressive strength

10. Determination ofshear strength Direct Shear Test.

Experimental Project/assignment/Micro Project: Students in groups of 4 to 6 will do the projects

1. Bringing soil samples from the field classify them by performing lab tests and then determining the optimum moisture content and maximum dry density.
2. Based on OMC and MDD they will prepare samples for determination of CBR.

Course Learning Outcome(CLO):
Upon completion of this course, the students will be able to:

1. Determine the index and engineering properties of soil
2. Evaluate the influence of water on engineering properties of soil
3. Evaluate the compressibility characteristics of soils in engineering practices
4. Determine the shear strength of soils by various methods

Text Books:

1. *Gopal Ranjan& A.S.R. Rao, Basic and Applied Soil mechanics, New Age Publisher, New Delhi (2016)*
2. *V.N.S. Murthy, A text book on Soil Mechanics and Foundation Engineering, U.B.S. Publisher, New Delhi.(2005)*
3. *Parshotham Raj, Geotechnical Engg., Pearson , New Delhi.(2013)*

Reference Books:

1. *Das B.M., Principles of Soil Mechanics, Thomson Publisher, USA. (2015)*
2. *Venkatramaiah Geotechnical Engg., New Age Publisher, New Delhi. (2012)*
3. *Singh Alam Modern Geotechnical Engineering, CBS Publishers, New Delhi.(2014)*

Evaluation Scheme:

Sr. No.	**Evaluation Elements**	**Weights (%)**
1.	MST	30
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30